Project 2

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Introduction

```
In the United States, there are elections when the two dominant political parties battle wit
biting victory and other times where one candidate wins by a landslide. For presidential ele
However, historically, linear regression methods have not been widely utilized for the purpo
call election. Due to a greater data size, elected officials in Congress will be considered 
performing models for Democratic versus Republican parties can be also compared.
```

Data

The dataset is curated from Kimberly Martin's analysis of voting trends on state legislation related to the transgender athletes ban in sports with one bill kept per state, spanning 2020-2023. There are 29 different states with related bills in total with each row representing individual congressional members. Martin tracks relevant bills based on the Movement Advancement Project (MAP) and the Human Rights Campaign (HRC). Though this dataset is originally intended to study trends specific to the transgender athlete ban in sports, it provides the most reliable and comprehensive demographic characteristics of elected officials. The complete dataset contains 52 columns and 3654 rows in total. There is at least one Democratic and one Republican senator or representative included for each of the 29 states. Key features include binary variables about representatives, such as gender, race, and sexual orientation. Additional information includes the proportions of religious, racial, age, marital status, and educational categories for each official's constituents.

Method

Programming Language

This analysis uses R Programming software and the following packages: dplyr, MASS, caret, glmnet, and tidymodels.

Data Cleaning

Prior to any modeling implementations, variables revealing election outcomes or irrelevant bill information will be omitted. We excluded any observations with a margin of victory equal to 1 as they are uncontested elections and 0 as these are inaccurate entries. We also dropped several rows with any omitted values that were most likely missing at random. After subsetting by party, this resulted in 700 observations for Democrats and 1760 for Republicans. Despite a smaller data size for Democrats, splitting this cohort will best serve our goal of observing prediction accuracy by party.

Full Model and Diagnostics

Through implementing multiple linear regression, our goal is to predict the margin of victory of a congressperson based on their personal demographic information and that of their constituents. The general formula for multiple linear regression is:

$$Y_{nx1} = X_{nxp}\beta_{px1} + \epsilon_{nx1}$$

Y represents the response vector while β is the coefficients associated with the X predictor matrix. ϵ corresponds to the noise in the data due to measurement error or true variations.

Response Variable

The response variable is the "margin of victory column" for each elected official's race. This ranges from 0.0148% to 99.6887%, with a median of 26.2647%, after removing uncontested elections.

Key Predictors

The following predictors remained after screening for variables that contained outcome information and athlete bill-specific columns.

```
 [1] "senate"          "lgbt"           "female"         "black"
 [5] "hispanic"        "nativeamasian"  "dist_votes"     "district_ideal"
 [9] "age25_44"        "age45_64"       "pctover65"      "pctwhite"
[13] "pctblack"        "pcthispanic"    "pctmarried"     "pcths"
[17] "pctbachelordeg" "rural"           "SSMC"           "EP"
```

- Senate (binary 0 or 1) refers to whether the congressperson is in the Senate (1) or House of Representatives (0).

- LGBT, Female, Black, Hispanic, NativeAmAsian are all binary variables that takes on the value 1 if this characteristic describes the official.
- "Dist_votes" is the total number of votes in the official's election while "district_ideal" is the political spectrum of the district on a scale of -1 (Very Liberal) to 0 (moderate) to 1 (Very Conservative).
- Age 25-44, Age 45-64, Over 65 represents the proportion of the district's constituents in that age group.
- [Pct=percentage] White, Black, Hispanic represents the proportion of the district's constituents in each racial group.
- [Pct] Married is the proportion of married individuals in the district
- [Pct] High school, bachelor degree is the proportion of the district who have completed these educational levels.
- Rural represents the proportion of the district considered rural as opposed to urban.
- SSMC is the number of same-sex marriage couples per 1000 households.
- EP is the number of evangelical protestants per 1000 constituents.

The full model includes all predictor variables and serves as a baseline comparison to subsets chosen by variable selection techniques. Data transformations were performed based on diagnostic plots using this comprehensive model.

For precise confidence intervals and accurate coefficient estimates, the following assumptions of the ordinary least squares linear regression should hold to guaranteed the best linear unbiased estimators (BLUE):

- Linearity: implies that a linear regression model is a good choice for representing the data (response Y vs. predictor(s) X)
- Independence of Errors: ensures accurate confidence intervals and p-values based reliable calculations of standard errors.
- Normality of Errors: allows for valid hypothesis tests and confidence intervals constructed
- Equal Variance of Errors: implies that the estimates for the coefficients are the most optimal, as standard errors will be accurate.

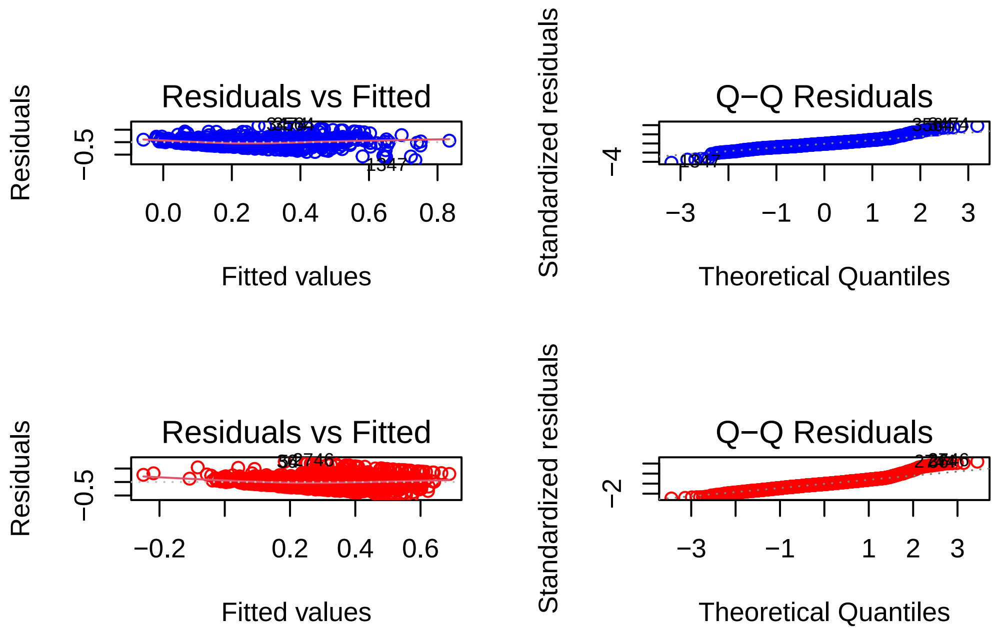
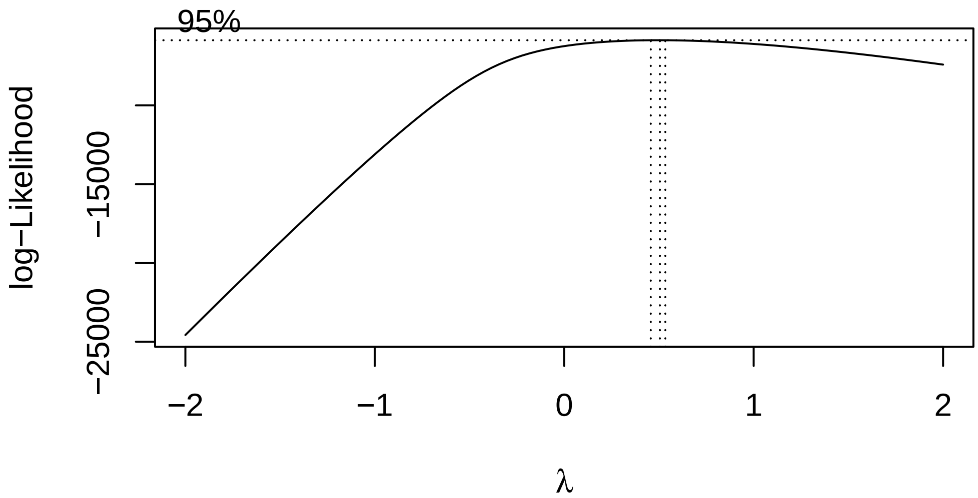
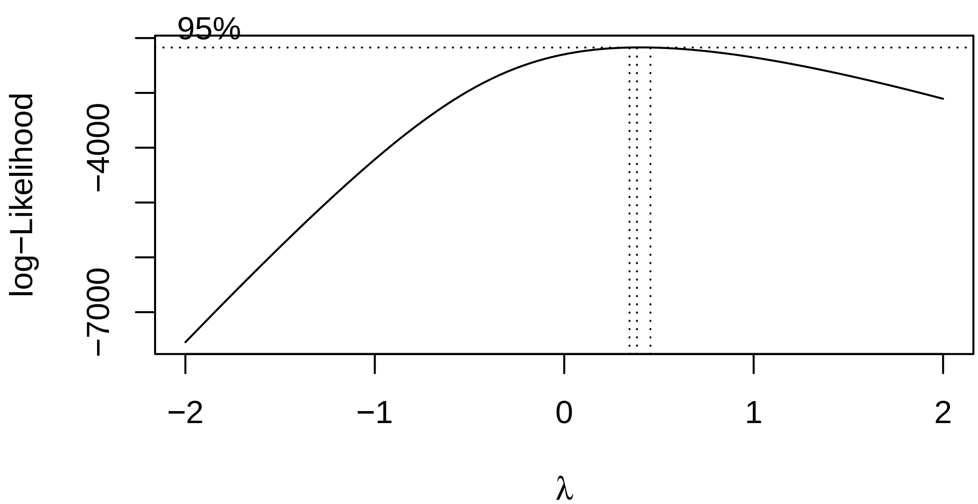


Figure 1: The left column represents the residual vs. fitted plot while the right covers the quantile-quantile plot, with blue for Democrat and red for Republican. Primarily in the Q-Q plot, the tail ends show deviation from the dotted line representing a perfect fit with the theoretical distribution. Both subsets violate the homoskedasticity assumption.

Data Transformations

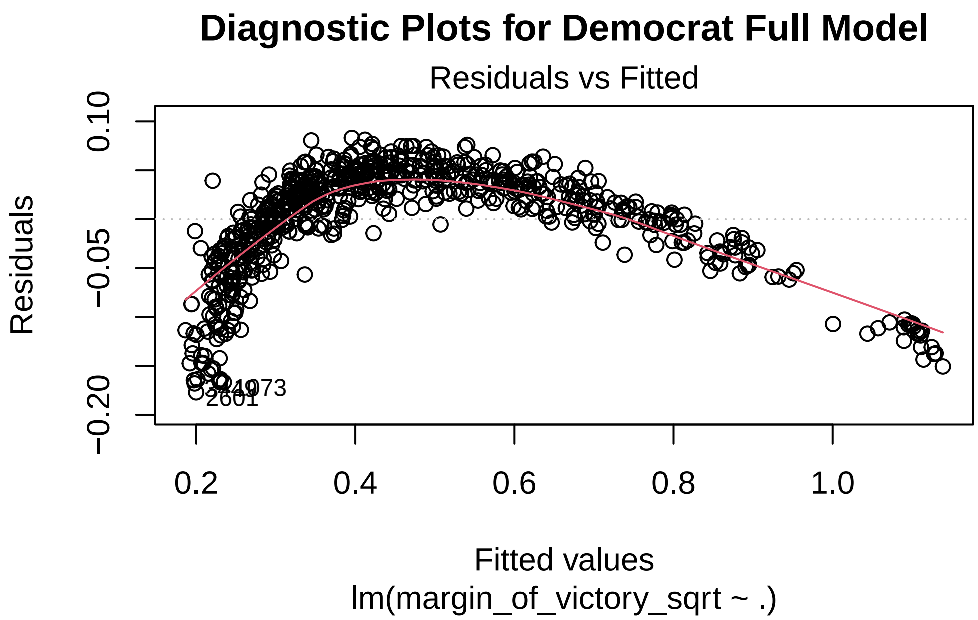
Due to violation of equal variance, shown clearly in the Q-Q plot, we will first transform the data using Box-Cox transformations. Based on commonly used λ parameters, we find that the square root transformation of the response variable is best for both the Democratic and Republican subsets.

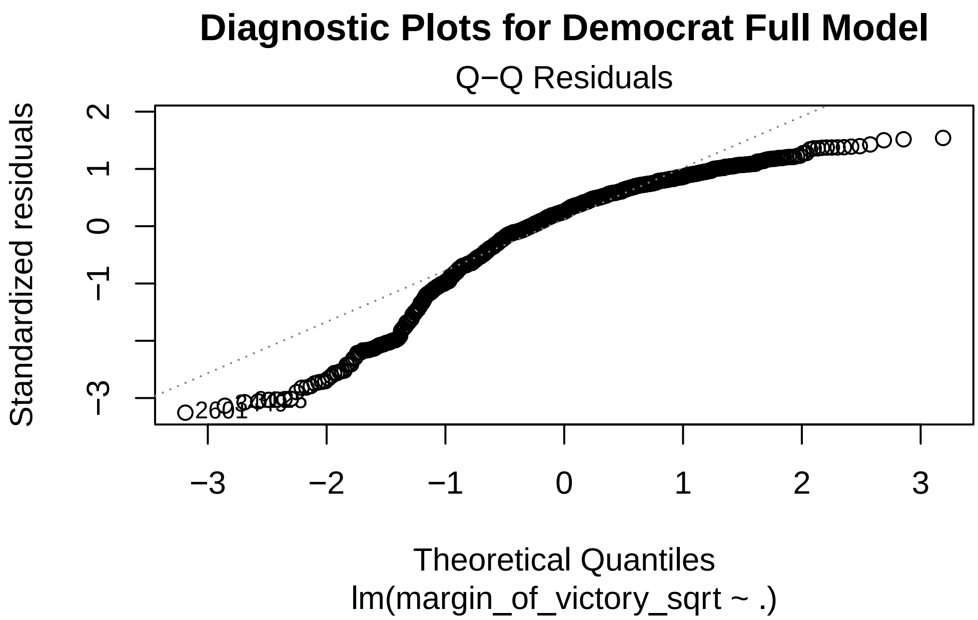


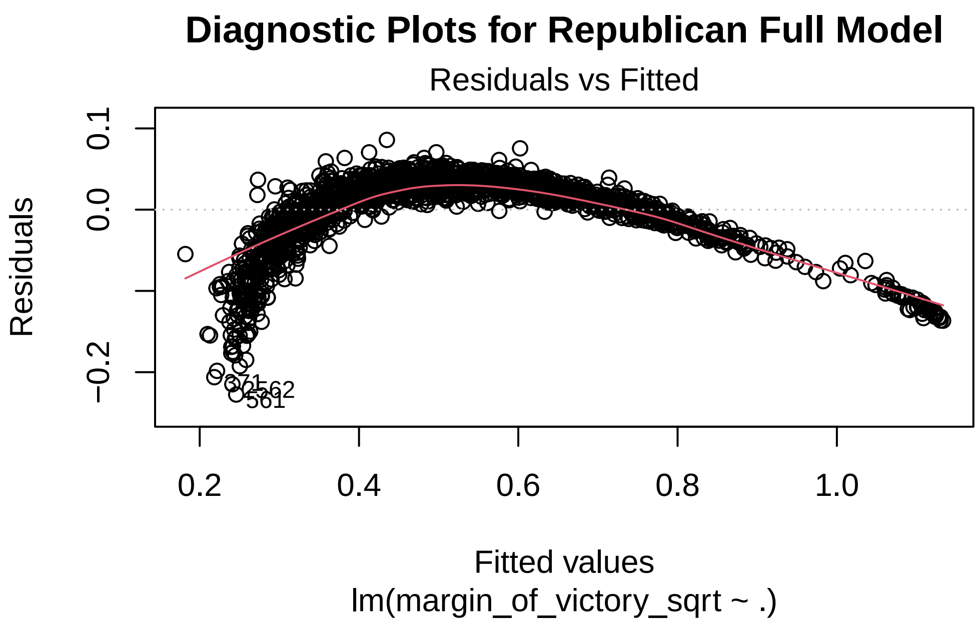


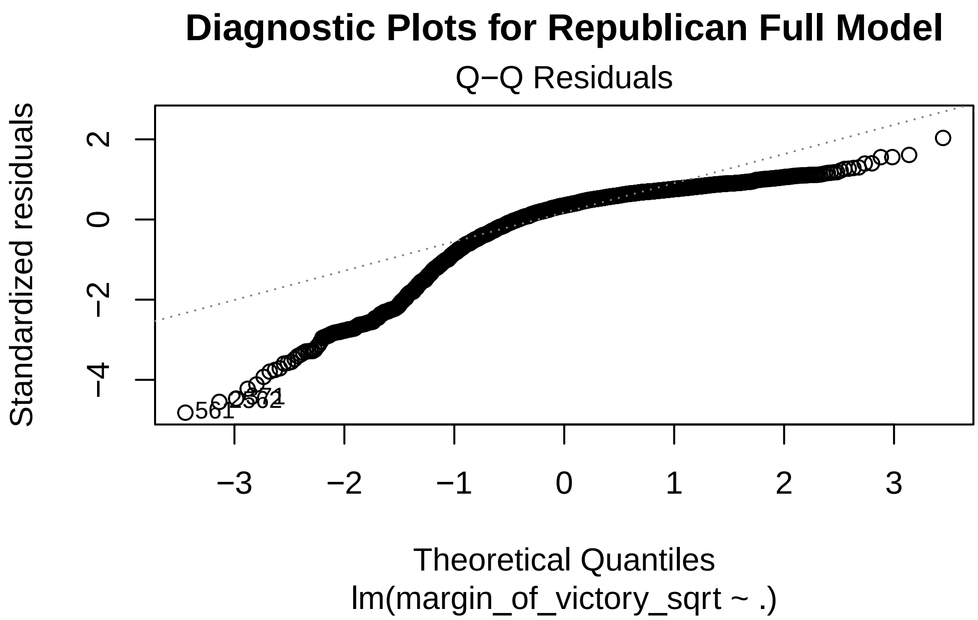

```
Attaching package: 'dplyr'

The following object is masked from 'package:MASS':
```

```
    select

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```




Diagnostic Plots for Democrat Full Model
Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(margin_of_victory_sqrt ~ .)
−3
−1
0
1
2
−3
−2
−1
0
1
2
3
2601

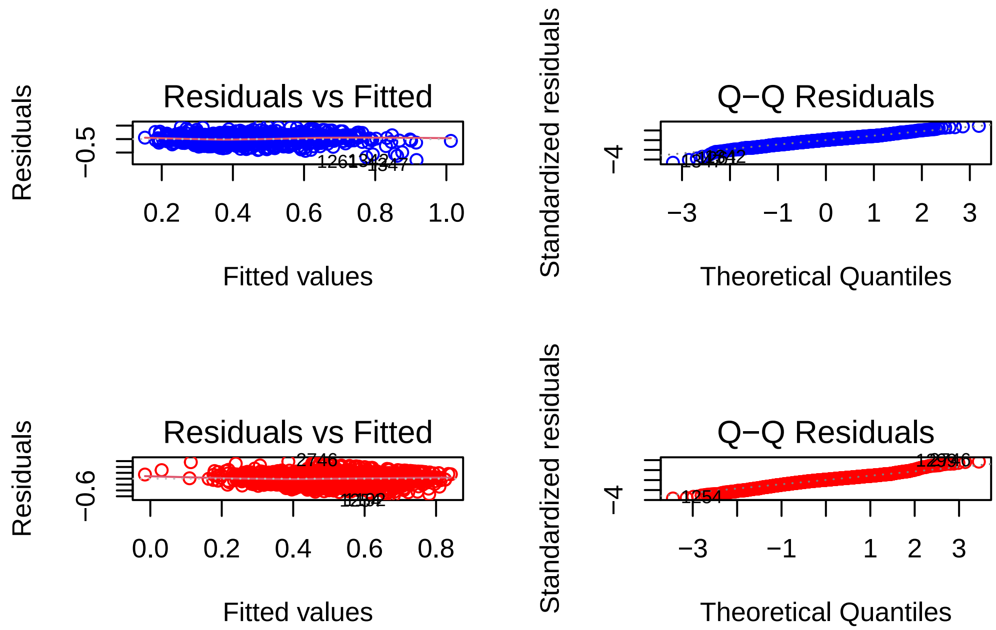

Diagnostic Plots for Republican Full Model
Residuals vs Fitted
Residuals
Fitted values
lm(margin_of_victory_sqrt ~ .)
0.1
0.0
−0.2
0.2
0.4
0.6
0.8
1.0
561



After this transformation, the diagnostic plot now shows a curve in the residuals vs. fitted plot, indicating that there are higher polynomial terms for some predictors for both Republicans and Democrats. We used an iterative loop to plot residuals against each predictor and add a colored LOWESS plot to highlight curvature. The predictors that showed some curvature for mainly improved for the Democratic subset after including B-splines terms of degrees of freedom 3. This is highlighted in (@fig) in the blue Q-Q plot, where the tail ends fall much more closely to the 45 degree line (representing a good fit with the theoretical distribution chosen).

The Republican subset requires more advanced exploration to resolve linear regression violations, but the analysis will proceed as is. Both subsets also revealed a segment with a linear pattern (upper right) in the residual vs. fitted plot, which could possibly be resolved using piecewise regression or adding interaction terms.

```
par(mfrow = c(2, 2))
plot(transformed_model_poly_dm, which = 1:2, col="blue")
plot(transformed_model_poly_rp, which = 1:2, col="red")
```




```r
par(mfrow = c(1, 1))
```

Splitting for Validation

With the limited data size in mind, each subset was split into a training/validation and a testing set with the ratio of 80/20. Cross-validation is performed to select the best model before observing performance on the test set.

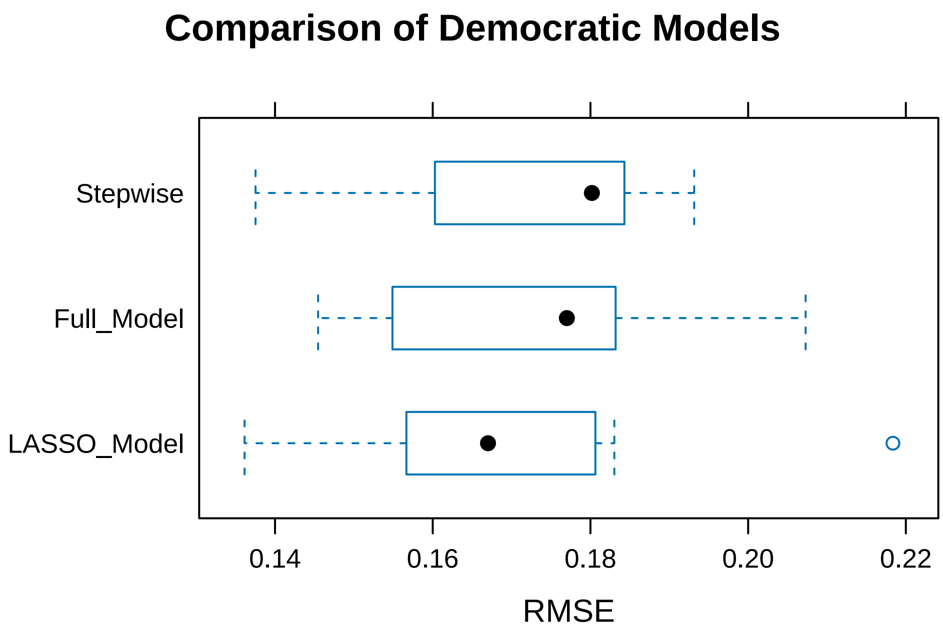
Model Comparisons (Variable Selection Techniques)

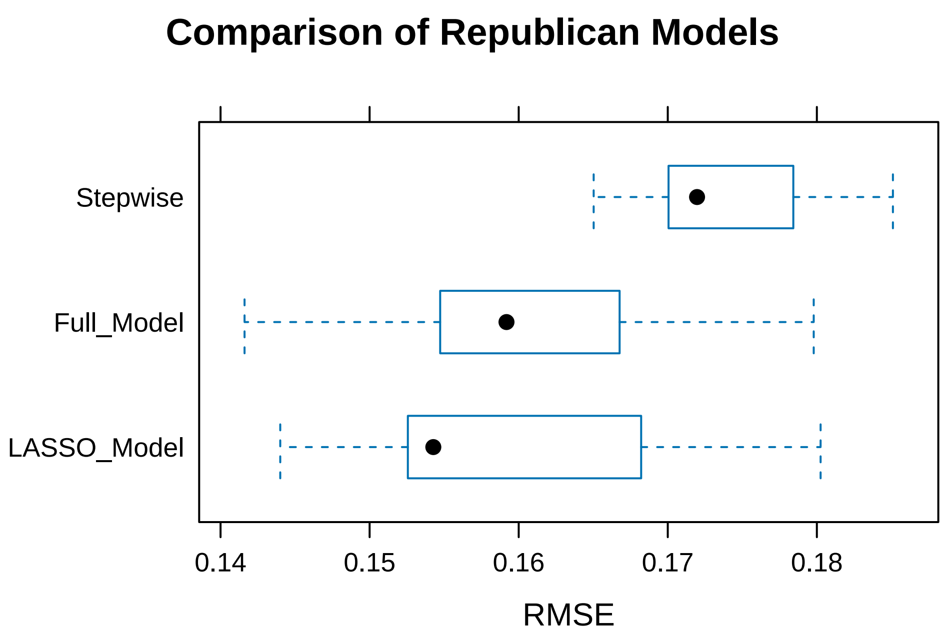
With the full model as the baseline, we used both LASSO and stepwise sequential search techniques to perform variable selection within each of the 10 folds of our training+validation set. The metric for selection was Root Mean Square Error (RMSE), which is the square root of the mean of differences between the predicted and observed values for the response variable.

LASSO is a regularization method that seeks to minimize a penalty term, $\lambda \sum |\beta_j|$. This shrinks the less important estimated coefficients to 0. For stepwise search, we fit a simple linear regression model for each of the $P-1$ predictors and add this predictor to the final model if it exceeds a predetermined threshold.

Result

Below are the boxplot comparisons of the results of the RMSE for the full, stepwise selected, and LASSO models:





Though each plot in @fig shows a wide, overlapping range of RMSE values, we will use the median of each. This is the LASSO model for Democrats and full model for Republicans.

Slope Interpretations

Democrat

Variables “zeroed” out by LASSO are considered not important for our response variable.

Key Variables:

- The estimated coefficient for “senate” is 0.01563, representing an expected increase of 0.01563 percentage points in the margin of victory for senators, when compared to those in the House of Representatives, holding all other variables constant.
- The estimated coefficient for “female” is -0.0148, representing an expected decrease of 0.0148 percentage points in the margin of victory for female officials, when compared to male, holding all other variables constant.
- The estimated coefficient for “age 45 to 64” is 0.0155, representing an expected increase of 0.0155 percentage points in the margin of victory given a 1 percentage point increase in constituents age 45 to 64, holding all other variables constant.
- The estimated coefficient for percentage “married” is -0.0743, representing an expected decrease of 0.0743 percentage points in the margin of victory given a 1 percentage point increase in constituents who are married, holding all other variables constant.
- The estimated coefficient for percentage “white” is -0.0501, representing an expected decrease of 0.0501 percentage points in the margin of victory given a 1 percentage point increase in constituents who are white, holding all other variables constant. The same interpretations follow for the other race categories.
- The estimated coefficient for percentage “bachelor degree” is 0.0596, representing an expected increase of 0.0596 percentage points in the margin of victory given a 1 percentage point increase in constituents who have obtained a bachelor degree, holding all other variables constant.

The same interpretations follow for the rest of the variables.

```r
best_lambda <- lasso_model_dm$bestTune$lambda
as.matrix(coef(lasso_model_dm$finalModel, s = best_lambda))
```

```
                        s=0.001
(Intercept)        0.4600275621
bs(SSMC, df = 3)1 -0.0055343631
```

```
bs(SSMC, df = 3)2  0.0035680300
bs(SSMC, df = 3)3 -0.0004334739
bs(EP, df = 3)1   -0.0182983066
bs(EP, df = 3)2    0.0000000000
bs(EP, df = 3)3    0.0000000000
senate             0.0156316931
lgbt               0.0000000000
female            -0.0148000925
black              0.0069436857
hispanic           0.0000000000
nativeamasian      0.0051518734
dist_votes        -0.0418587863
district_ideal    -0.0477358925
age25_44           0.0137498405
age45_64           0.0155033759
pctover65          0.0124611486
pctwhite          -0.0500904191
pctblack           0.0306978681
pcthispanic        0.0289258161
pctmarried        -0.0743766888
pcths             -0.0188511633
pctbachelordeg     0.0596919648
rural              0.0073489137
SSMC               0.0000000000
EP                -0.0142845650
```

Republican

Key Variables:

- The estimated coefficient for "lgbt" is -1.601e-01, representing an expected decrease of 1.601e-01 percentage points in the margin of victory for officials part of the LGBTQ+ community, when compared to those who are not, holding all other variables constant.
- The estimated coefficient for "district ideology" is 1.871e-01, representing an expected increase of 1.871e-01 percentage points in the margin of victory for senators, given a 1 point increase on the political spectrum (moving towards conservative end), holding all other variables constant.
- The estimated coefficient for "age 45 to 64" is -8.020e-03, representing an expected decrease of 8.020e-03 percentage points in the margin of victory given a 1 percentage point increase in constituents age 45 to 64, holding all other variables constant.

- The estimated coefficient for percentage "married" is 8.943e-03, representing an expected increase of 8.943e-03 percentage points in the margin of victory given a 1 percentage point increase in constituents who are married, holding all other variables constant.
- The estimated coefficient for percentage "white" is 3.370e-03, representing an expected increase of 3.370e-03 percentage points in the margin of victory given a 1 percentage point increase in constituents who are white, holding all other variables constant. The same interpretations follow for the other race categories.

The same interpretations follow for the rest of the variables (note: Evangelical Protestants and same-sex married couples are both singular, which is reasonable due to religious beliefs opposing SSMC; proportion with a bachelor degree or in rural living are also singular).

```r
summary(model_full_rp)
```

```
Call:
lm(formula = .outcome ~ ., data = dat)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.58143 -0.08796  0.00932  0.09430  0.58294 

Coefficients: (4 not defined because of singularities)
                                Estimate Std. Error t value Pr(>|t|)    
(Intercept)                    3.610e-01  1.733e-01   2.083  0.03743 *  
`bs(pctbachelordeg, df = 3)1`  1.683e-02  8.559e-02   0.197  0.84414    
`bs(pctbachelordeg, df = 3)2` -2.962e-01  5.654e-02  -5.238 1.82e-07 ***
`bs(pctbachelordeg, df = 3)3` -2.352e-01  8.390e-02  -2.803  0.00512 ** 
`bs(rural, df = 3)1`          -4.207e-02  3.319e-02  -1.268  0.20507    
`bs(rural, df = 3)2`           8.455e-02  3.123e-02   2.707  0.00685 ** 
`bs(rural, df = 3)3`          -3.607e-02  2.274e-02  -1.586  0.11290    
`bs(SSMC, df = 3)1`           -9.712e-02  6.019e-02  -1.614  0.10681    
`bs(SSMC, df = 3)2`           -1.948e-01  1.603e-01  -1.215  0.22435    
`bs(SSMC, df = 3)3`           -1.211e-01  1.610e-01  -0.752  0.45223    
`bs(EP, df = 3)1`             -1.122e-01  6.266e-02  -1.790  0.07364 .  
`bs(EP, df = 3)2`              4.153e-01  7.205e-02   5.765 9.67e-09 ***
`bs(EP, df = 3)3`             -1.766e-01  1.024e-01  -1.725  0.08477 .  
SSMC                                  NA         NA      NA       NA    
EP                                    NA         NA      NA       NA    
senate                        -1.217e-02  8.990e-03  -1.354  0.17593    
lgbt                          -1.601e-01  7.153e-02  -2.238  0.02537 *  
female                         7.920e-03  9.891e-03   0.801  0.42338    
```

```
black                          1.789e-02  4.432e-02   0.404  0.68654    
hispanic                       2.993e-02  3.308e-02   0.905  0.36570    
nativeamasian                 -9.489e-02  5.663e-02  -1.676  0.09398 .  
dist_votes                    -4.509e-08  8.880e-08  -0.508  0.61170    
district_ideal                 1.871e-01  2.035e-02   9.195  < 2e-16 ***
age25_44                      -6.344e-03  1.989e-03  -3.190  0.00145 ** 
age45_64                      -8.020e-03  1.703e-03  -4.709 2.69e-06 ***
pctover65                     -8.772e-04  1.365e-03  -0.643  0.52050    
pctwhite                       3.370e-03  7.946e-04   4.241 2.35e-05 ***
pctblack                      -1.172e-03  9.115e-04  -1.286  0.19865    
pcthispanic                    2.305e-03  1.010e-03   2.281  0.02268 *  
pctmarried                     8.943e-03  8.629e-04  10.364  < 2e-16 ***
pcths                         -1.875e-03  1.693e-03  -1.107  0.26826    
pctbachelordeg                        NA         NA      NA       NA    
rural                                 NA         NA      NA       NA    
`SSMC:EP`                      9.118e-05  1.728e-04   0.528  0.59791    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1583 on 1730 degrees of freedom
Multiple R-squared:  0.3896,	Adjusted R-squared:  0.3793 
F-statistic: 38.07 on 29 and 1730 DF,  p-value: < 2.2e-16
```

Testing Results

Subset	Best Model	Validation Median RMSE	Test RMSE	Test R-Squared
Republican	Full	0.1603866	0.1625	0.4517
Democratic	LASSO	0.1651067	0.1757	0.4904

Discussion

Ultimately, the best models and Democratic and Republican elected officials are different, with the Democratic model having a higher R^2.

References